

JEE MAIN REVISION MASTER: MOTION IN A STRAIGHT LINE

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 121 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Motion In A Straight Line - Core Application	121 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Motion In A Straight Line - Core Application (Very High Yield — 121 PYQs)

Core Principle: Key problem pattern in Motion In A Straight Line.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "A gas balloon is going up with a constant velocity of 10m/s . When this balloon reached a height of 75 m , a stone is dropped from it and ..." [Ans: D]
- [2026 Morning Shift] Q#1 (MCQ): "Water drops fall from a tap on the floor, 5 m below, at regular intervals of time, the first drop strikes the floor when the sixth drop begi..." [Ans: C]
- [2026 Evening Shift] Q#2 (MCQ): "The velocity (v) versus time (t) plot of a particle is shown in the figure, for a time interval of 40 s . The total distance travelled b..." [Ans: C]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 8th April Evening Shift (Q#1)

A gas balloon is going up with a constant velocity of 10 m/s . When this balloon reached a height of 75 m , a stone is dropped from it and balloon keeps moving up with the same velocity. The height of the balloon when the stone hits the ground is ____ m. (Take $g=10\text{ m/s}^2$)

- (A) 85
- (B) 150
- (C) 129
- (D) 125 [Correct]

Correct Answer: D

Solution Analysis: When an object is dropped from a moving carrier (like the balloon), it gets the velocity of that carrier at the exact moment of release. Initial velocity of the stone (balloon) is $u_s=10\text{ m/s}$ (upwards). Initial height of the stone is $h=75\text{ m}$. The acceleration due to gravity is $g=-10\text{ m/s}^2$ (downwards). Using the second equation of motion for the stone's displacement along vertical direction, $y=u_y t + \frac{1}{2} a_y t^2$. Since the stone eventually hits the ground (at height 0), its displacement from the starting point (75 m) is $y = -75\text{ m}$. $-75 = (10)t + \frac{1}{2}(-10)t^2$